

Networking Track
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eBPF & Switch Abstractions

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- The Multi-Host NIC Abstraction
- The Switchdev Based Multi-Host NIC Abstraction
- Currently upstream (as of 2 weeks ago)
  - Boot
  - Setting Switchdev Mode
  - Loading Qdiscs
- Next Steps (currently being upstreamed)
  - Generalising Qdisc Offload
  - Adding clsact Qdiscs (u32, cls_bpf)
- Future Work

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Background: HW, BPF JIT, NIC as a Switch

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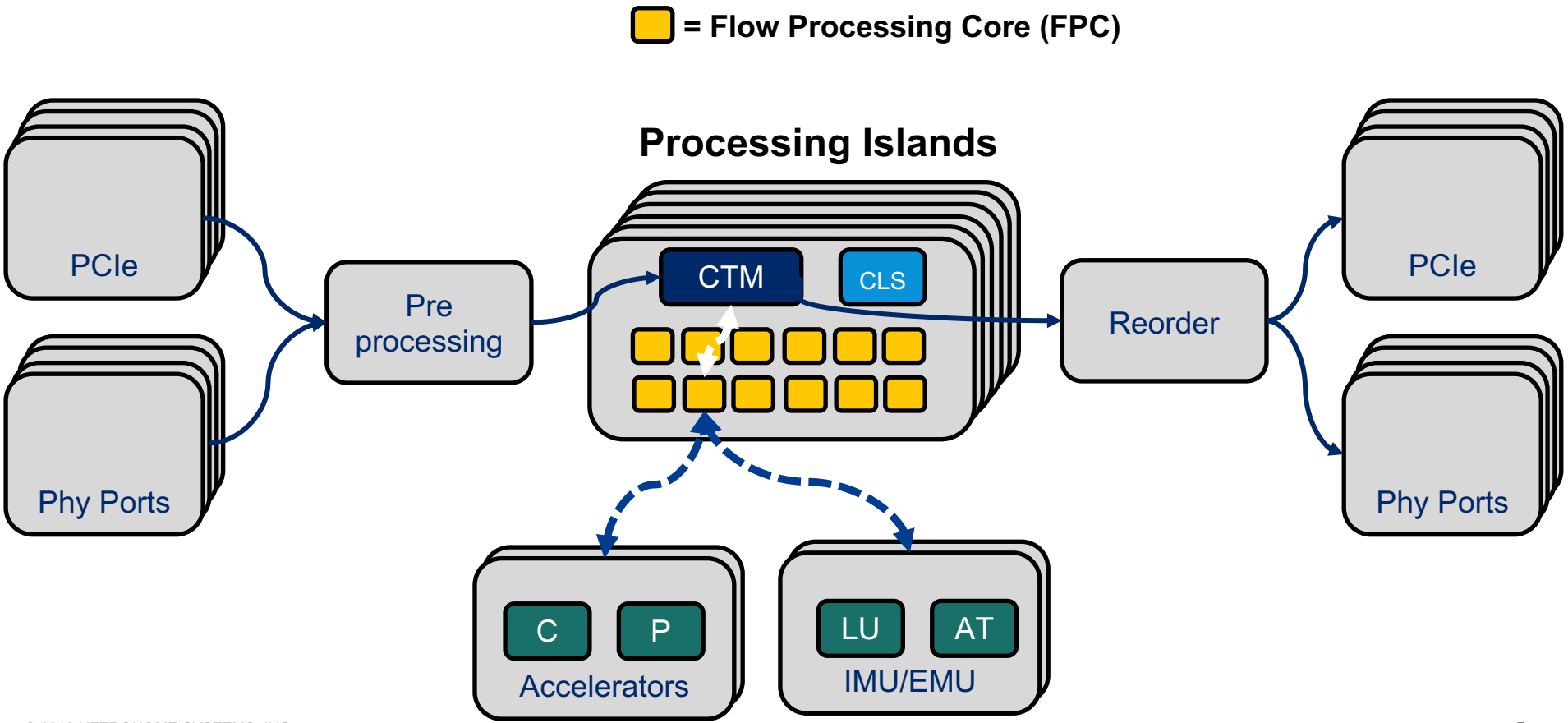
Background: HW

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- Many core, fully programmable network processor
  - 48-96 preprocessing cores
  - 54-120 programmable cores, 8 threads per core, MIMT
  - up to 4 PCIe
  - Up to 40 ports supported
  - ~17MB of on chip memory
  - 2-24GB of DRAM
  - Distributed & Transactional memory architecture
- Low power
  - ~ 10-35W (dependent on chip + frequency)

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Background: HW Datapath

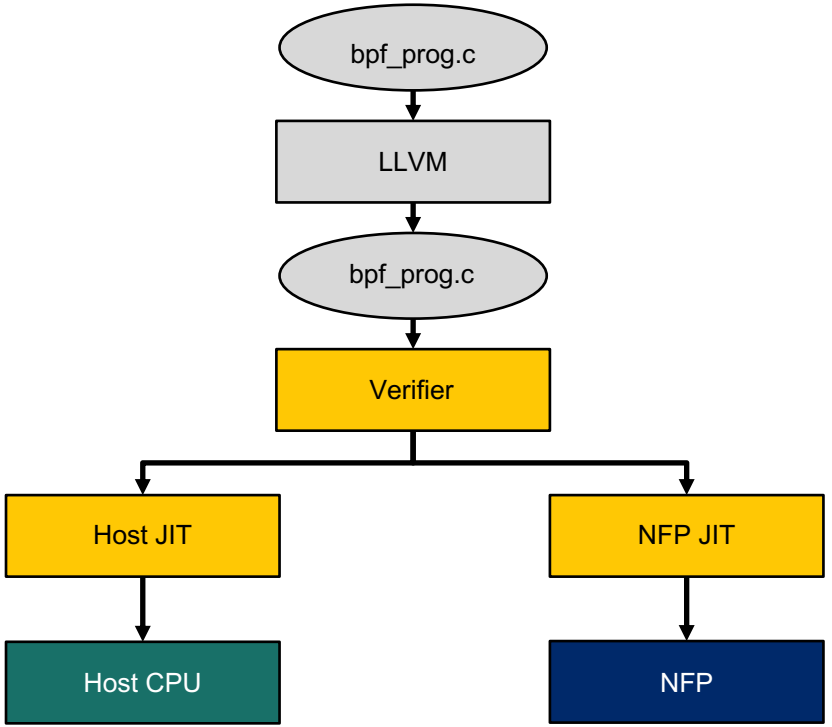


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Background: eBPF Offload Architecture

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- Program is written in standard manner
- LLVM compiled as normal
- The nfp's jit is called like any other architectures jit
- This converts the BPF bytecode to NFP machine code
- Translation reuses the verifier infrastructure in kernel
- Defining the FPC datapath code using BPF

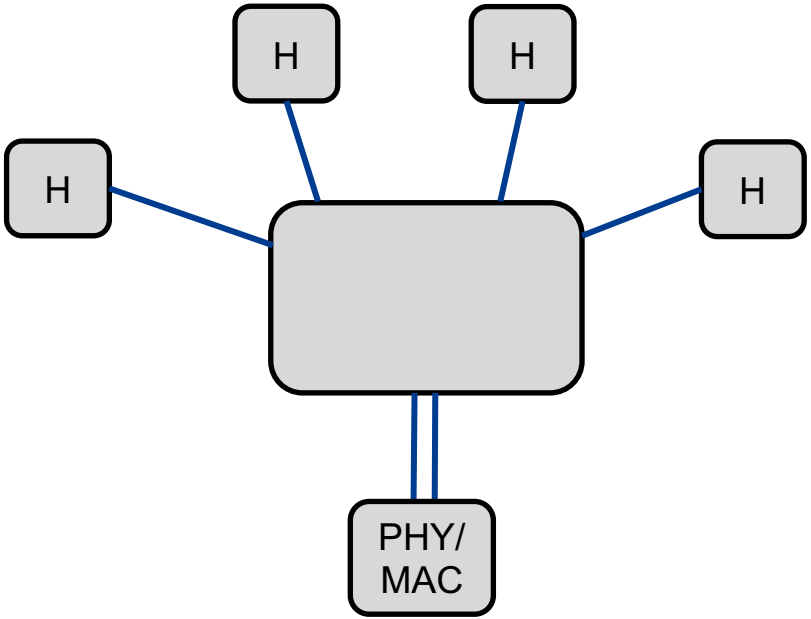


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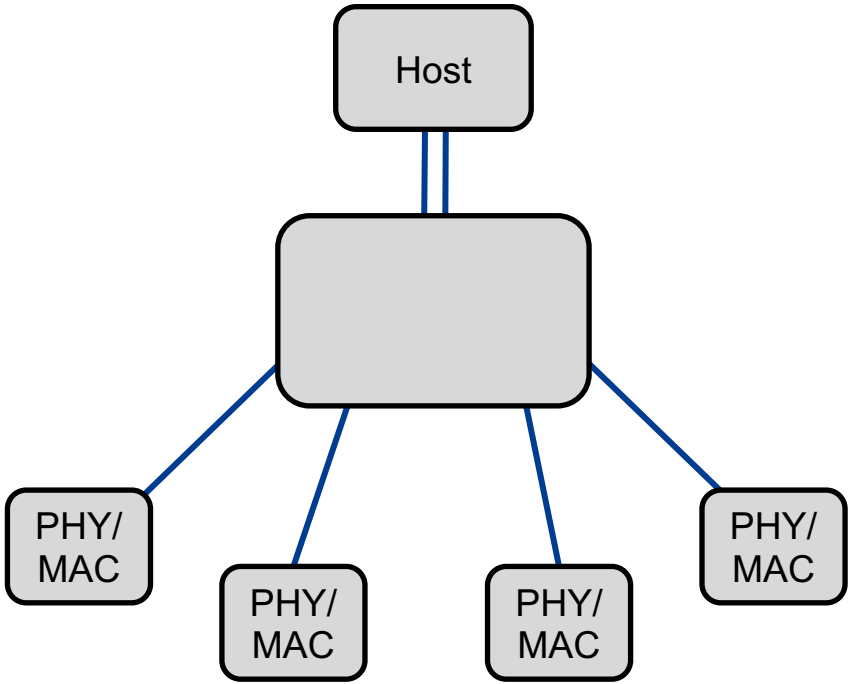
Background: NIC as a Switch

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Multi-Host



“Multi-Homed”



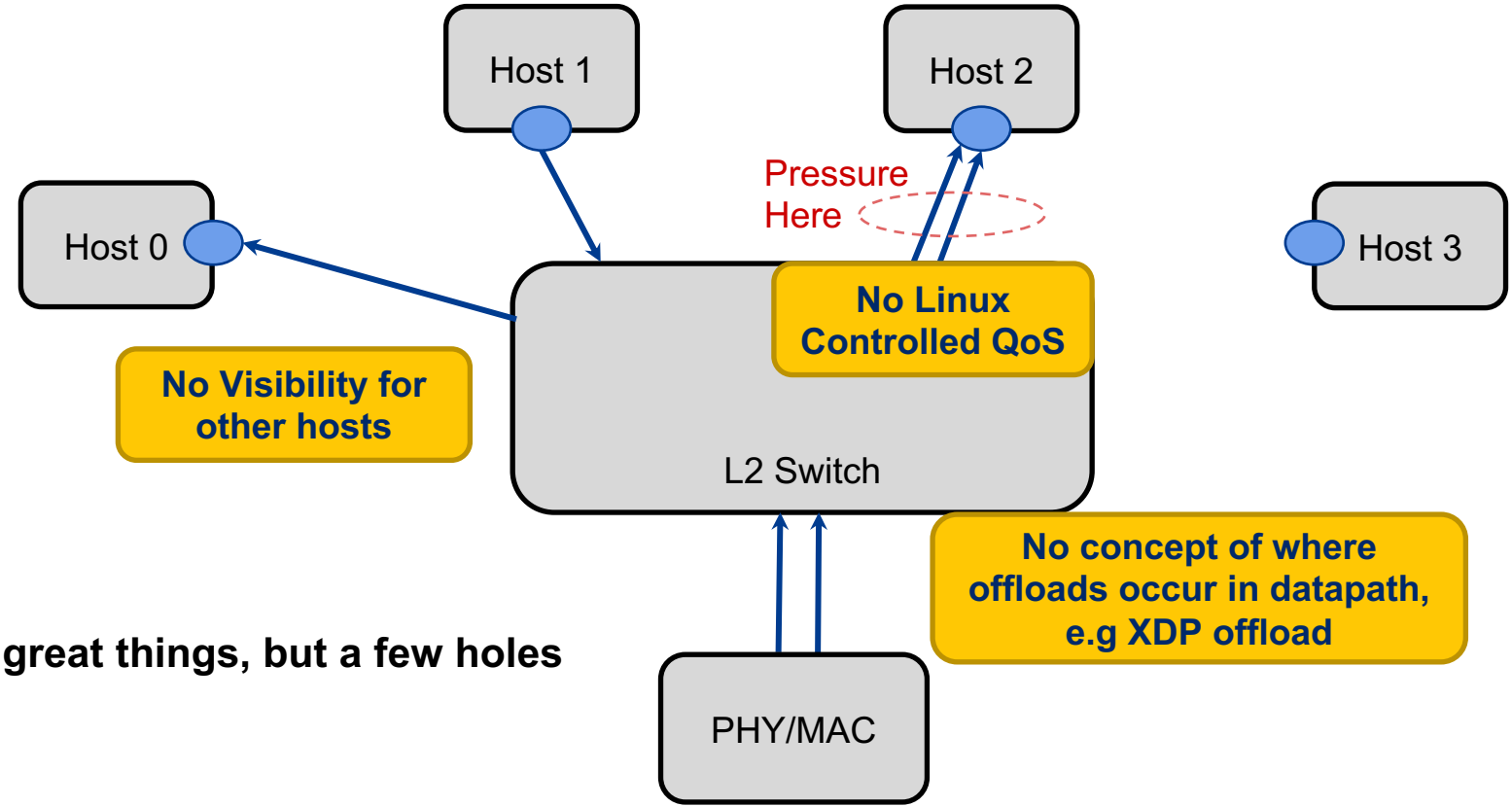
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Multi-Host NIC Architecture

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Multi-Host NIC Abstraction: Today

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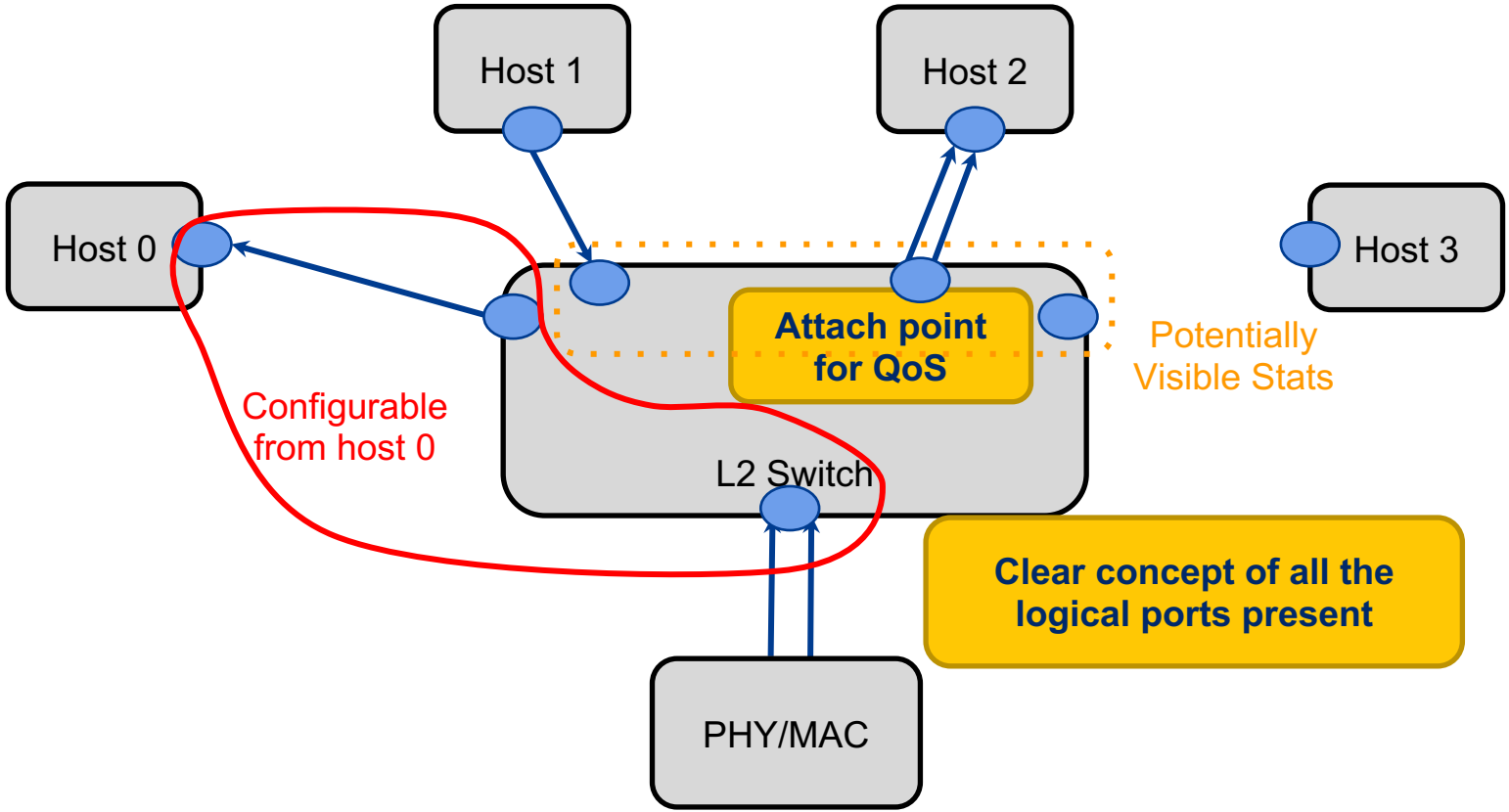


Many great things, but a few holes

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Multi-Host NIC Abstraction: Switchdev Based

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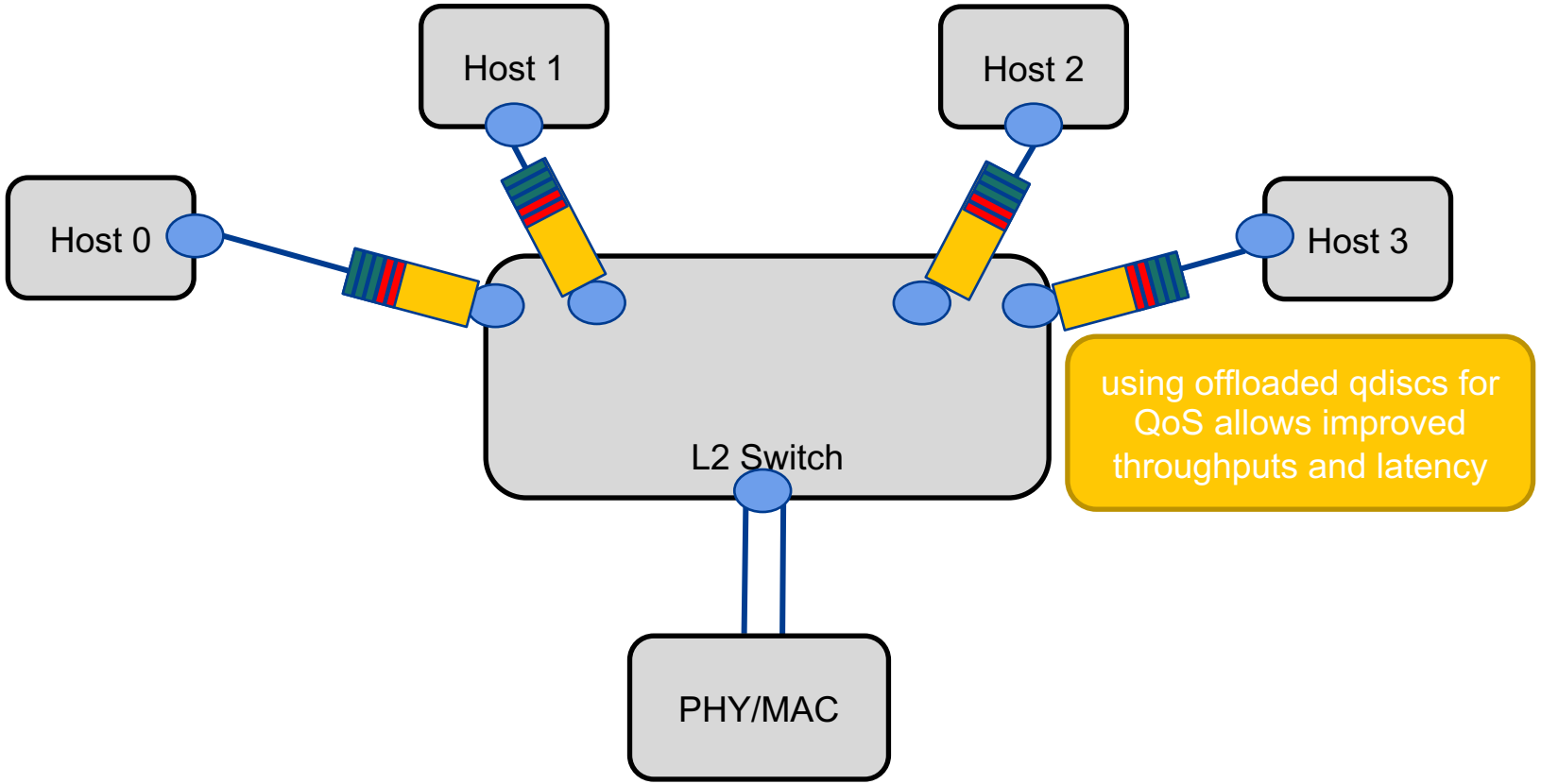
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Current Work: Switchdev and Simple Qdisc Offload

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Multi-Host NIC Abstraction: Incorporating Qdiscs

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3 Key Steps

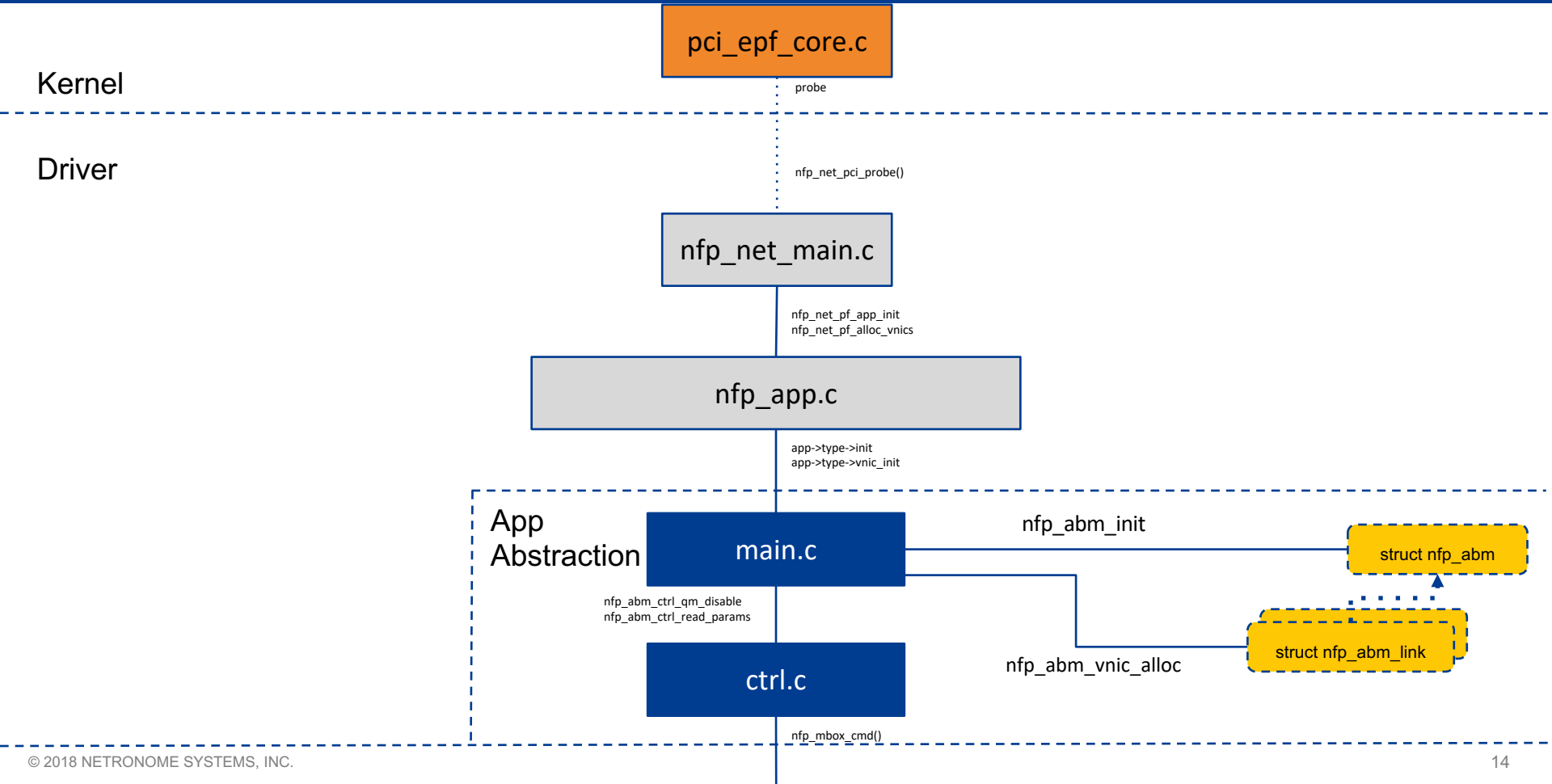
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- Initialisation (.probe)
  - App Initialisation
  - vNIC Allocation
- Entering Switchdev Mode (.devlink_eswitch_set)
  - Spawning Representors
- Qdisc Setup (.ndo_tc_setup)
  - Attaching Qdisc to struct nfp_abm_link

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Initialisation

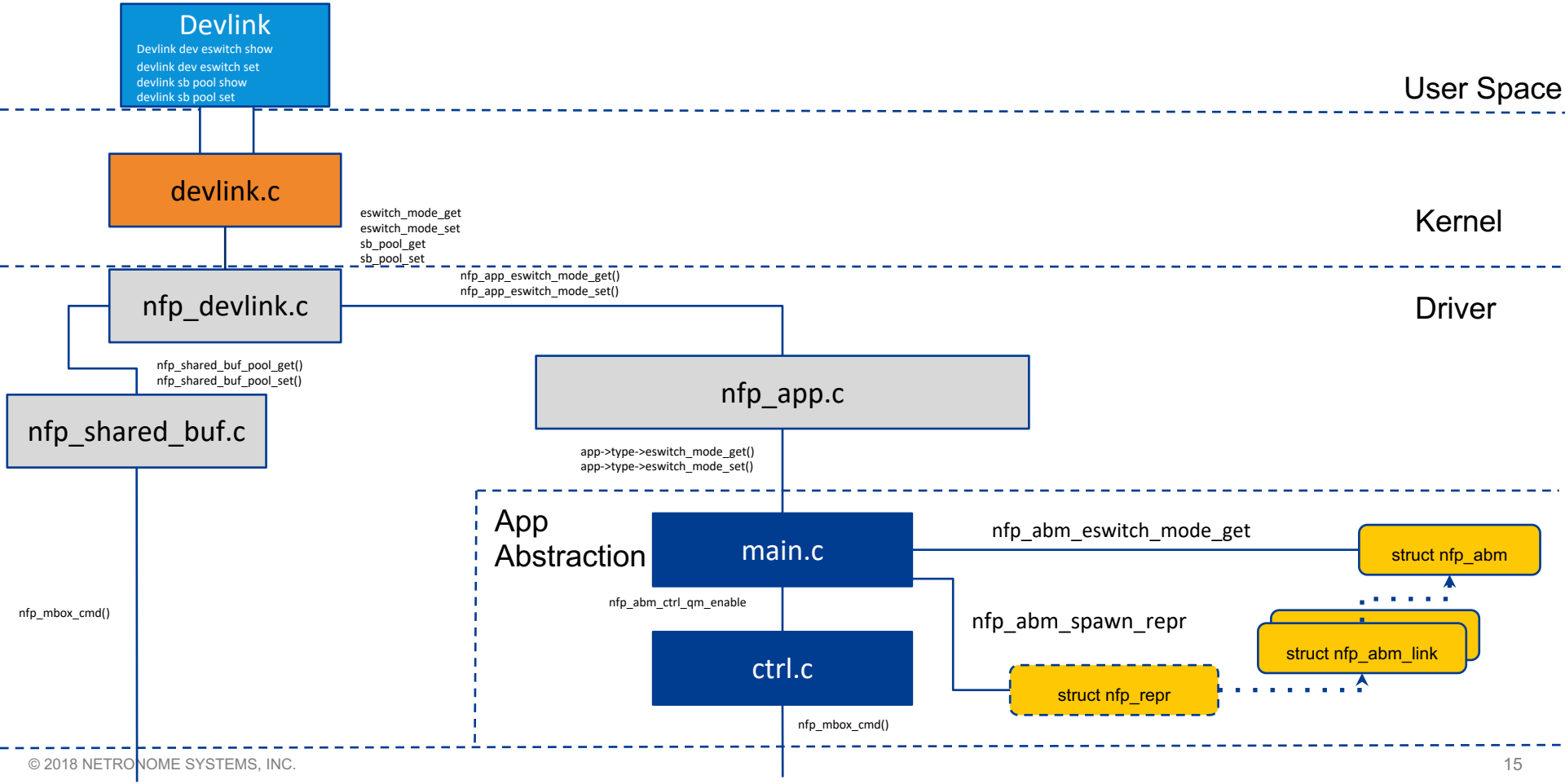
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Entering Switchdev Mode

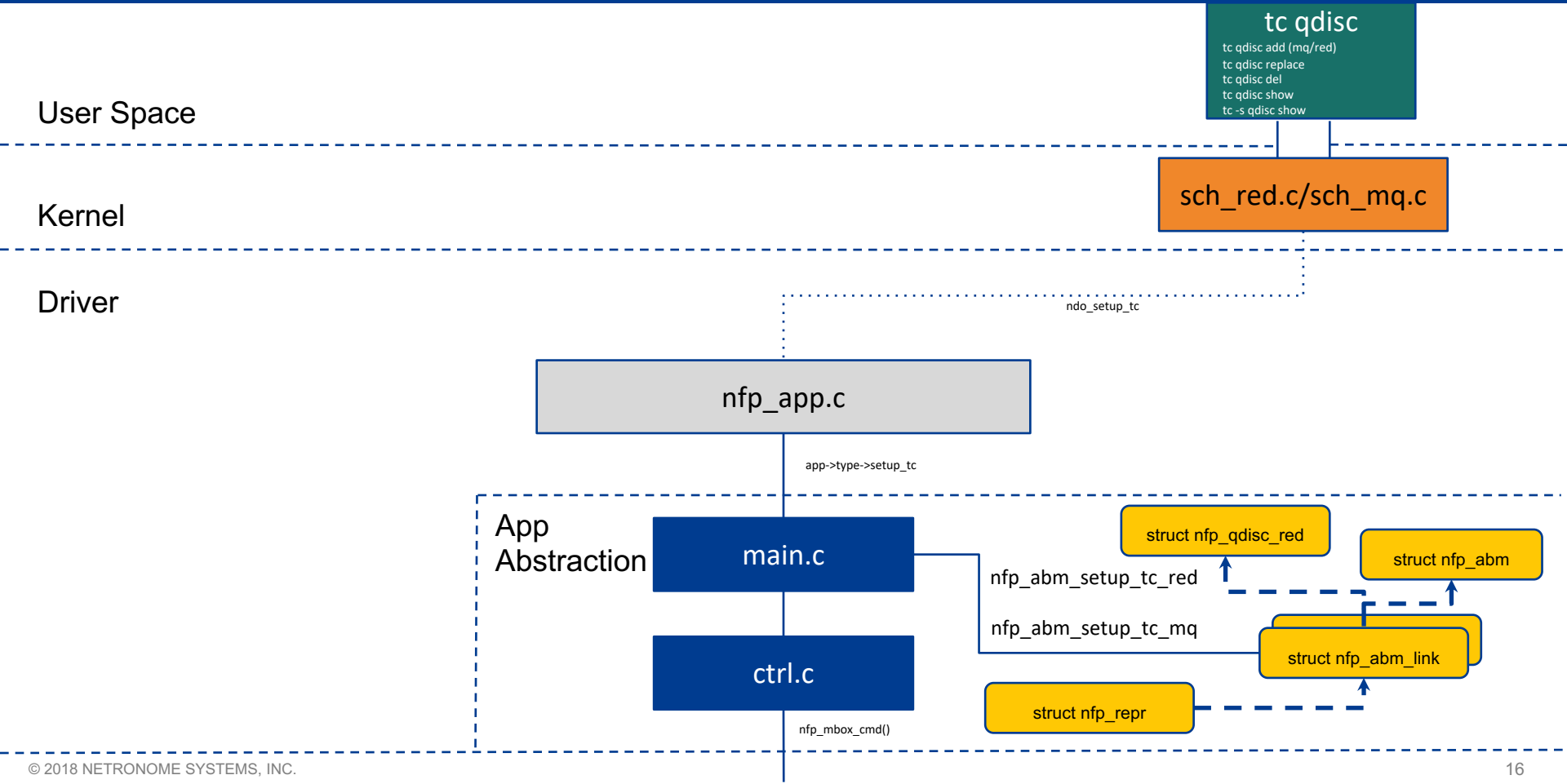
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Qdisc Offload

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Next Steps: Extending the Egress Representor Architecture

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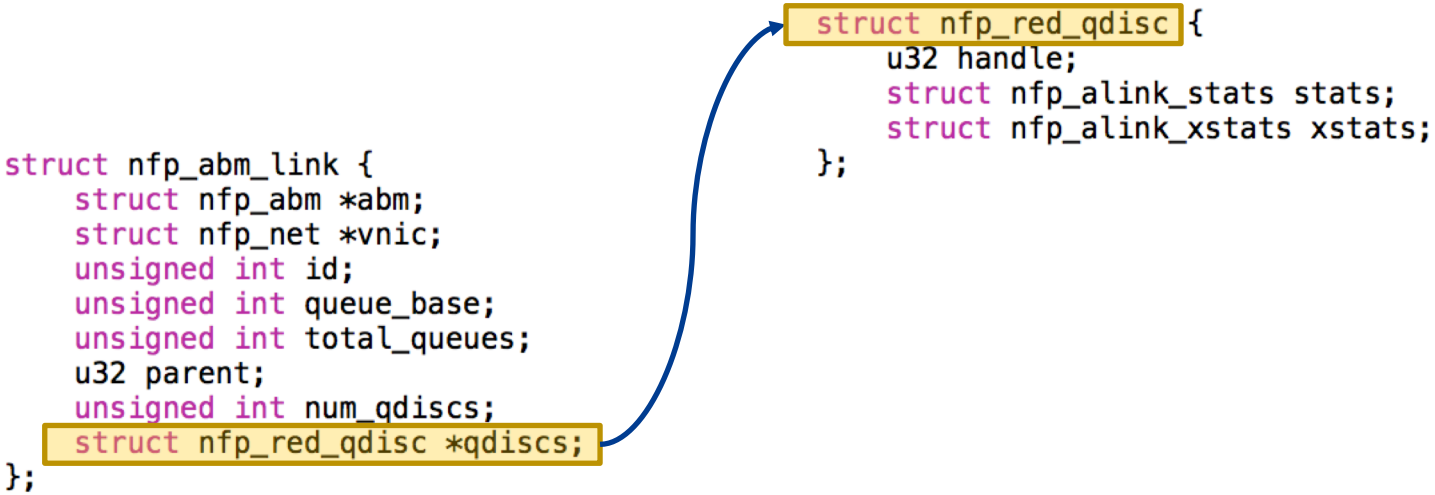
Items Covered

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- Generalising Qdisc Offload
  - Structure changes
    - nfp_abm_link
    - nfp_qdisc
- The clsact Qdisc
  - Motivation
  - Architecture

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Before



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After

```
struct nfp_abm_link {
    struct nfp_abm *abm;
    struct nfp_net *vnic;
    unsigned int id;
    unsigned int queue_base;
    unsigned int total_queues;
    struct nfp_qdisc *root_qdisc;
    struct radix_tree_root qdiscs;
};
```

```
struct nfp_qdisc {
    struct net_device *netdev;
    enum nfp_qdisc_type type;
/***/
    struct nfp_qdisc **children;

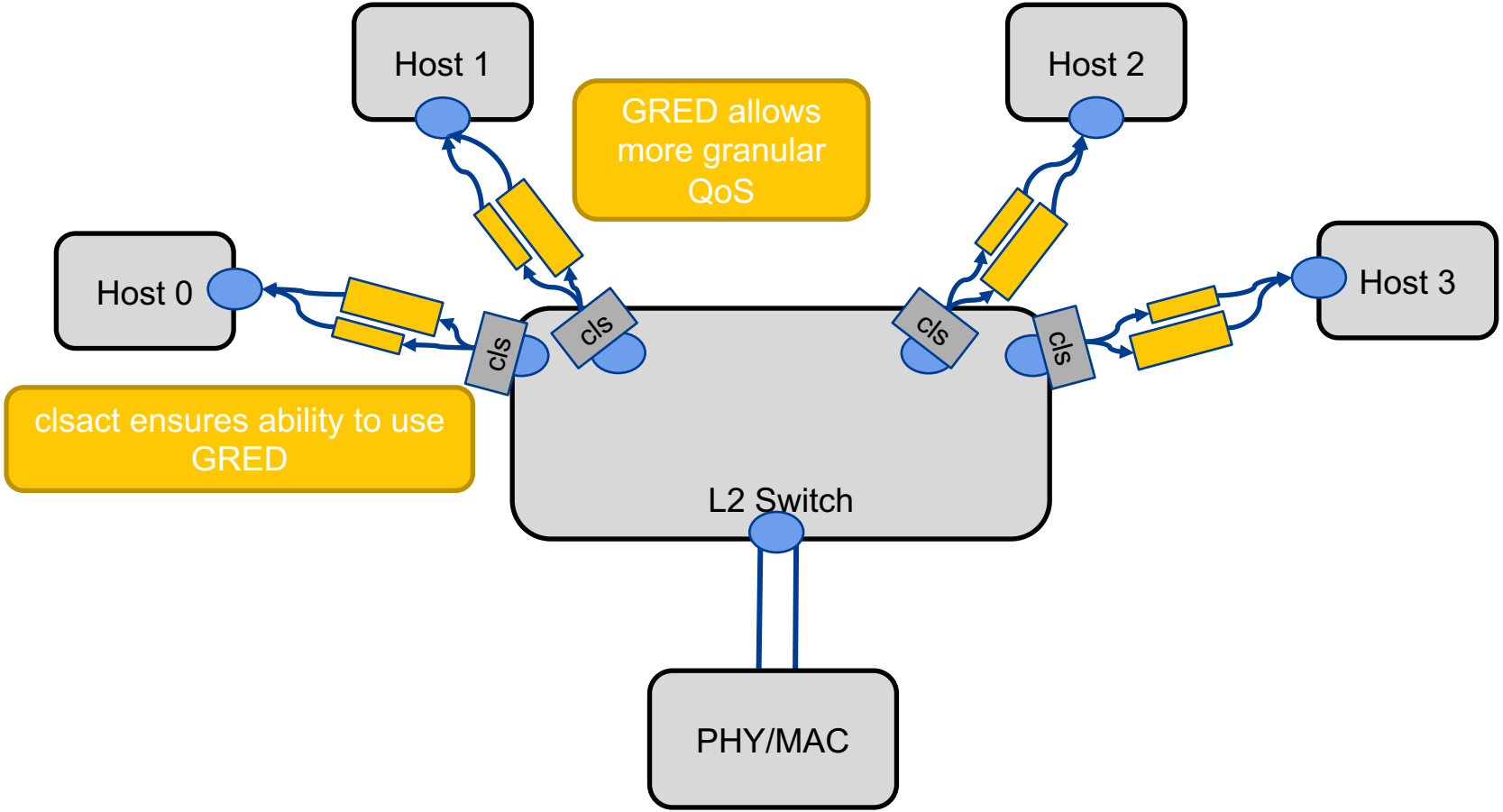
/***/

    union {
        /* TC_SETUP_QDISC_PRIO */
        struct {
            const u32 *map;
        } prio;
        /* TC_SETUP_QDISC_RED */
        struct {
            u32 threshold;
            struct nfp_alink_xstats stats;
            struct nfp_alink_xstats prev_stats;
        } red;
    };
};
```

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clsact Qdisc: Motivation

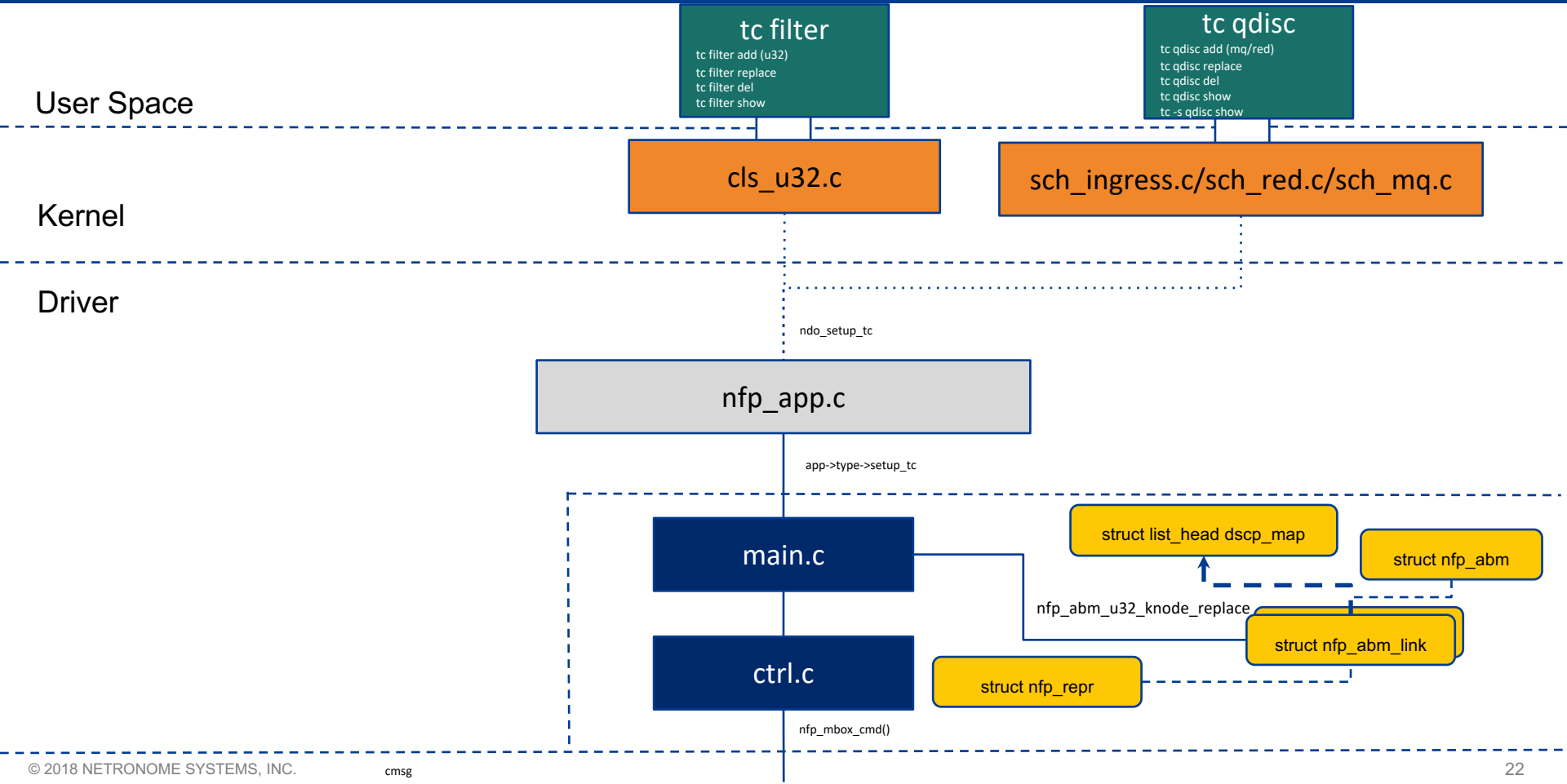
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clsact Qdisc: Architecture

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Future Work: Multi-host BPF Offload

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Items Covered

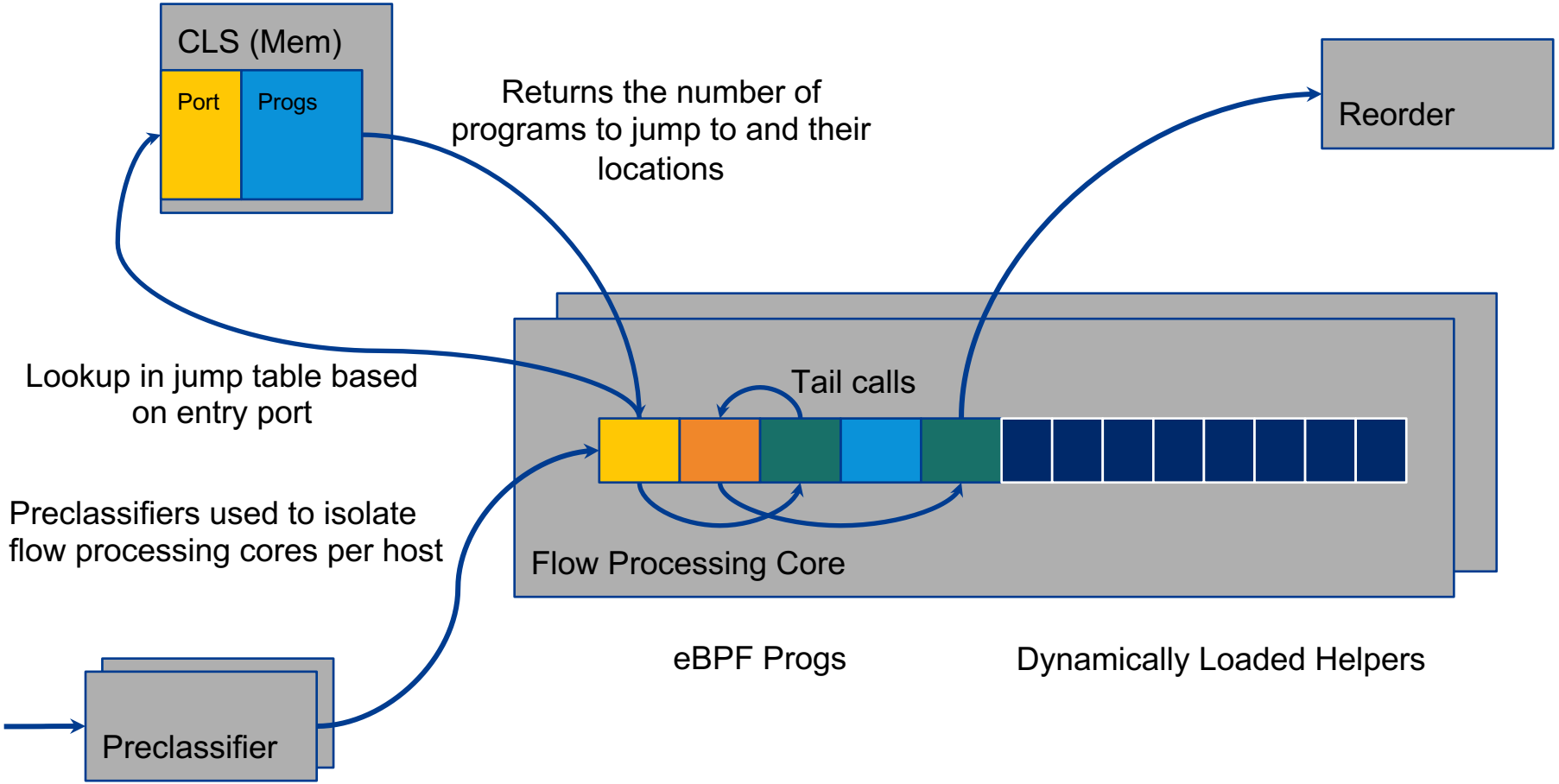
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- Firmware and BPF JIT
  - Flow Processing Core Datapath
- cls_bpf and Switchdev
  - Architecture
- XDP for Multihost Systems
  - Problems
  - Proposed Abstraction

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Firmware and JIT Changes

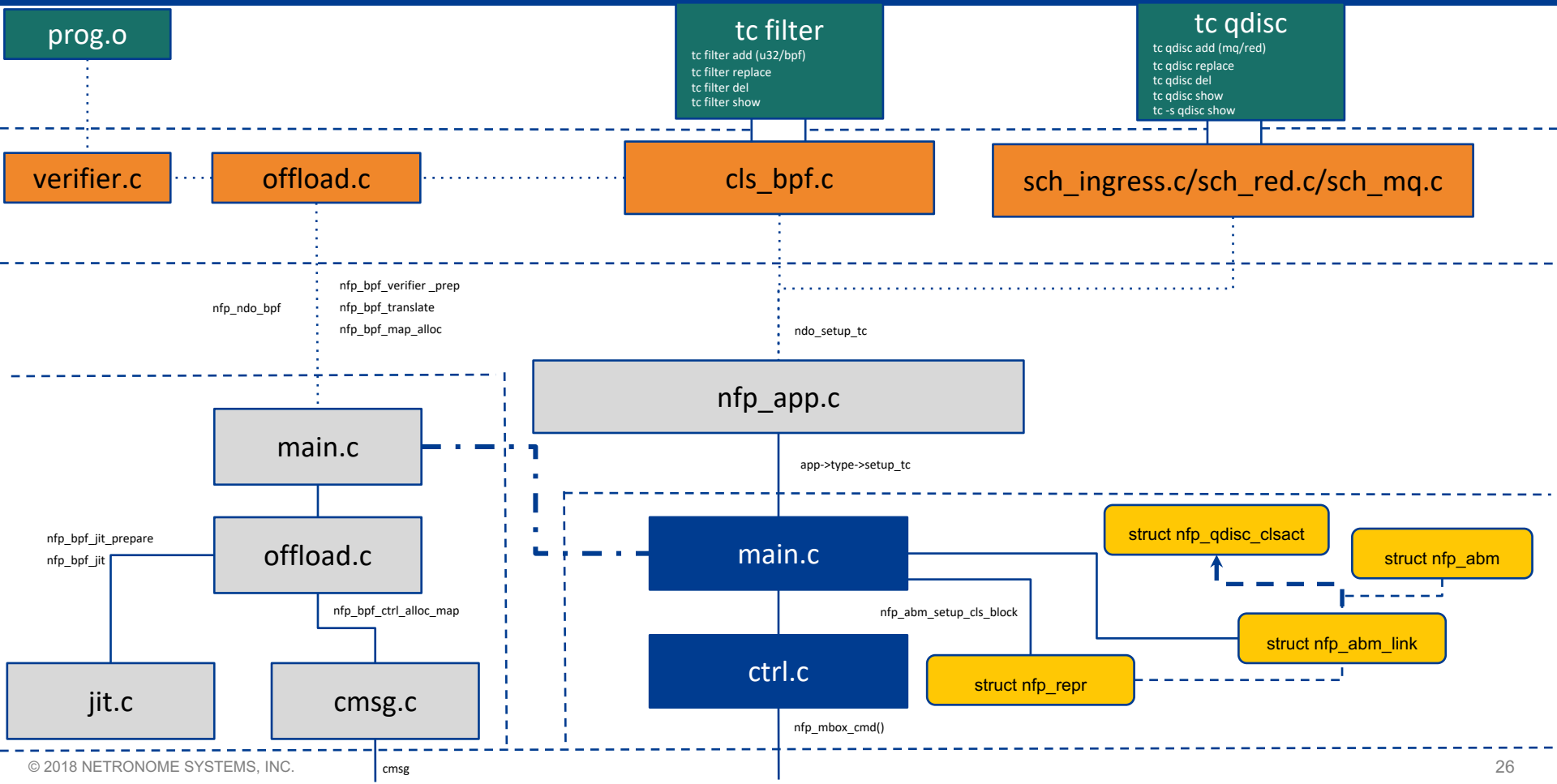
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cls_bpf on Egress

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XDP on Multi-Host systems

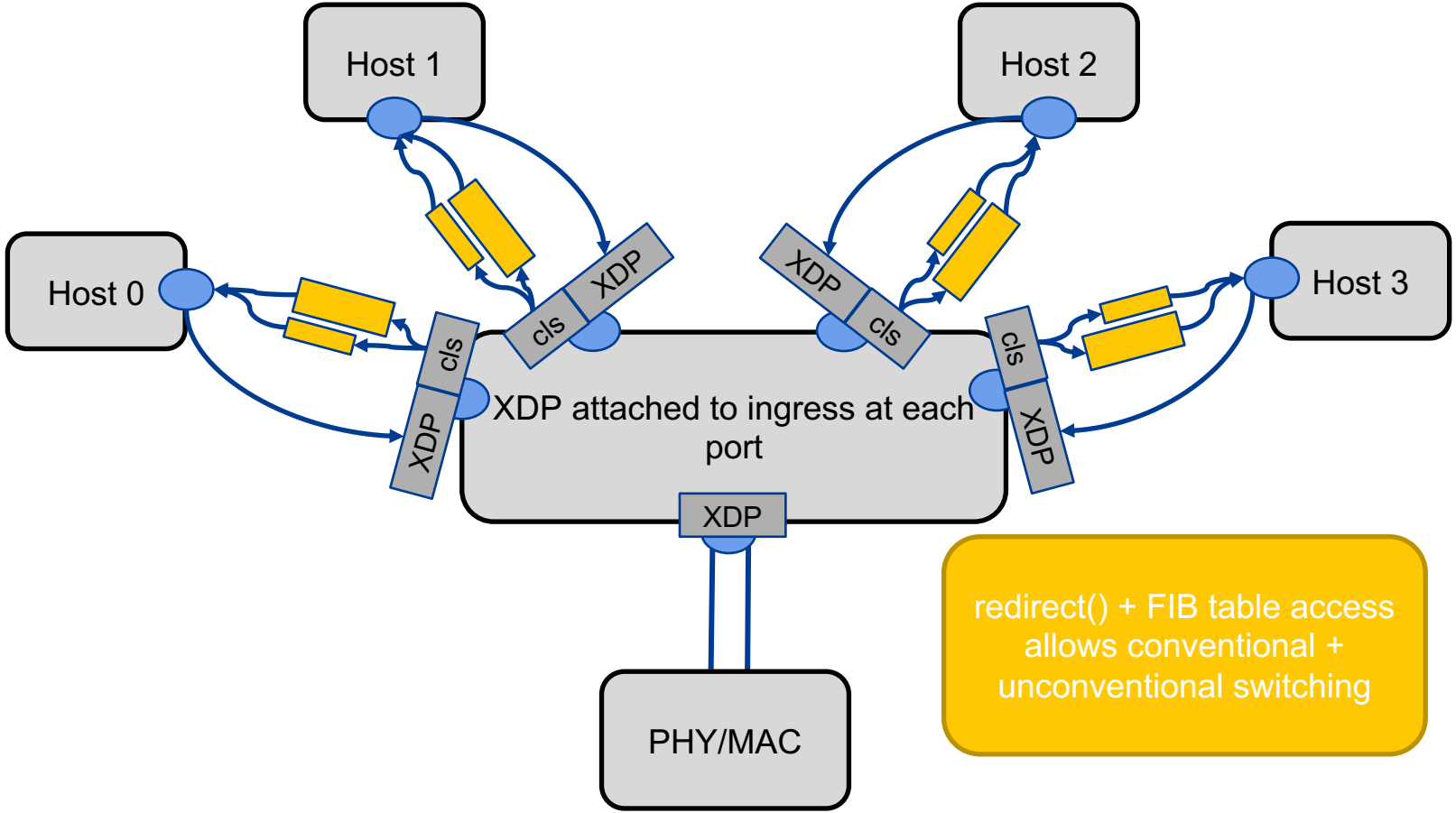
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- Challenges that have to be solved
  - XDP is an RX exclusive hook
  - Heterogenous architecture support is nascent
  - Security
- However more and more of the potential problems are falling away
  - e.g David Ahern's recent work on exposing the FIB table

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XDP on Multi-Host systems: Proposed Abstraction

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Summary

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- Proposing a fully flexible datapath for a multi-host NIC
- Achieved through a combination of switchdev, qdisc offload, cls_bpf and XDP
- Work in progress
  - Switchdev architecture and qdisc offload has been upstreamed
  - Next is simple clsact support
  - Followed by cls_bpf & XDP
- Provides potential for BPF defined pipelines in heterogenous architectures
  - See Jakub's talk at the microconference for more!

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Credit (Team)

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- Jakub Kicinski
- Jiong Wang
- Quentin Monnet
- David Beckett
- Edwin Peer
- Johan Moraal
- Mary Pham

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Thank you!



Discussion

Questions/Comments

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